The Numerical Framework:

A Damping Model, the Riemann Zeta Zeros, and the Proposed Integer Cutoff $x_p = 21$

A Conceptual and Exploratory Study within the Al-Hindara Framework

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Abstract

This paper presents a conceptual numerical framework within the Al-Hindara project. It introduces the normalization proposal $Z + C + A = 1$ and studies a candidate damping function involving the integer $x_p = 21$, defined as the floor of the ordinate of the second positive nontrivial zero of the Riemann zeta function.

The proposed function is

$$Z(x) = x^{-5} \ln(x) \sin\left(\frac{2\pi}{x}\right) e^{-x/x_p}, \qquad x > 1.$$

The choice $x_p = 21$ is a modeling decision motivated by the numerical value $\gamma_2 \approx 21.022039$. It is not a consequence of a theorem about the zeta function.

For the accumulation function

$$A(x) = \int_1^x Z(t)\, dt,$$

sign analysis shows that A decreases on $1 < x < 2$ and increases on $x > 2$. Consequently, the proposed function does not have a maximum at $x = 42$ on the interval $1 < x \leq 200$. This distinction between a numerical proposal and a verified mathematical property is made explicit.

The paper also discusses the quantity $K = 2\pi e^{-1/42}$ and its proposed role in the model. No asymptotic connection between this quantity and the Riemann zeta function is established here.

A companion study of tree crown ratios is discussed as an exploratory empirical investigation. Its small sample does not establish a causal or universal biological relationship. The paper therefore separates exact mathematical results, reported empirical observations, and conjectural modeling proposals.

Keywords: Riemann zeta function, nontrivial zeros, damping function, integer cutoff, Al-Hindara, numerical modeling, reciprocal Silver Ratio, exploratory analysis.

1 Introduction

The Al-Hindara project proposes the conceptual normalization relation

$$Z + C + A = 1. \tag{1}$$

In this paper, Equation (1) is treated as a conceptual framework rather than an established mathematical identity. A rigorous quantitative interpretation would require explicit definitions of Z, C, and A, their domains, their units where applicable, and the normalization procedure that makes their sum equal to one.

The present paper investigates a candidate numerical function with an exponential damping term. The selected damping parameter is

$$x_p = 21.$$

This value is motivated by the integer part of the second positive nontrivial zero ordinate of the Riemann zeta function.

The objectives are:

1. Define the numerical cutoff x_p precisely.
2. Introduce and analyze the proposed damping function.
3. Examine the mathematical behavior of its accumulation function.
4. Distinguish numerical observations from proved results.
5. Describe the proposed relationship with the companion tree-shadow study without asserting an unproved causal link.

The paper does not claim to prove the Riemann Hypothesis, derive the proposed function from the zeta function, or establish a universal law connecting number theory and biological morphology.

2 The Riemann Zero and the Proposed Integer Cutoff

Let $\zeta(s)$ denote the Riemann zeta function. The first positive ordinates of its nontrivial zeros on the critical line are conventionally denoted by $\gamma_1, \gamma_2, \ldots$, where a zero is written as

$$s = \frac{1}{2} + i\gamma.$$

The first two positive ordinates are approximately

$$\gamma_1 \approx 14.134725, \qquad \gamma_2 \approx 21.022039.$$

See Edwards [1] for background on the Riemann zeta function and its zeros.

We define the candidate integer cutoff by

$$x_p := \lfloor \gamma_2 \rfloor = 21. \tag{2}$$

Within the Al-Hindara project, the integer 21 is referred to as the *Yemeni Constant*. This is a project-specific name for a model parameter, not a recognized mathematical constant established by this paper.

The choice in Equation (2) is a modeling convention. Taking the floor of γ_2 gives 21, but this operation alone does not imply that the resulting integer has a special analytic role in the distribution of zeta zeros.

Any such role would require a separate mathematical argument or reproducible numerical evidence against an appropriate comparison model.

3 Definition of the Damping Function

For $x > 1$, define

$$Z(x) = x^{-5} \ln(x) \sin\left(\frac{2\pi}{x}\right) e^{-x/x_p}, \qquad x_p = 21. \tag{3}$$

Equivalently,

$$Z(x) = x^{-5} \ln(x) \sin\left(\frac{2\pi}{x}\right) e^{-x/21}. \tag{4}$$

The expression contains four main factors:

- x^{-5}: an algebraic factor that decreases with x.
- $\ln(x)$: a logarithmic factor.
- $\sin(2\pi/x)$: an oscillatory factor whose argument depends on the reciprocal of x.
- $e^{-x/21}$: an exponential damping factor.

The sine term should not be described as a sinusoid with a constant period in x. Its argument is $2\pi/x$, so the oscillation is not periodic with a fixed period on the x-axis.

Equation (3) defines a mathematical function. Its interpretation as a physical field, a biological model, or a quantity derived from the Riemann zeta function has not been established.

4 Sign and Asymptotic Behavior

For $1 < x < 2$, the angle $2\pi/x$ lies between π and 2π. Therefore,

$$\sin\left(\frac{2\pi}{x}\right) < 0.$$

Since $x^{-5} > 0$, $\ln(x) > 0$, and $e^{-x/21} > 0$ for $x > 1$, it follows that

$$Z(x) < 0, \qquad 1 < x < 2. \tag{5}$$

At $x = 2$,

$$Z(2) = 0. \tag{6}$$

For $x > 2$, the angle $2\pi/x$ lies between 0 and π, so

$$Z(x) > 0, \qquad x > 2. \tag{7}$$

These sign properties follow directly from the definition and do not depend on numerical simulation.

As $x \to \infty$, the small-angle approximation gives

$$\sin\left(\frac{2\pi}{x}\right) \sim \frac{2\pi}{x}.$$

Consequently,

$$Z(x) \sim x^{-5} \ln(x) \frac{2\pi}{x} e^{-x/21} \tag{8}$$

$$= 2\pi x^{-6} \ln(x) e^{-x/21}. \tag{9}$$

Thus,

$$Z(x) \sim 2\pi x^{-6} \ln(x) e^{-x/21}, \qquad x \to \infty. \tag{10}$$

This is an asymptotic statement about the defined function $Z(x)$. It is not an asymptotic formula for the Riemann zeta function or its zeros.

5 The Accumulation Function

Define the continuous accumulation function

$$A(x) = \int_1^x Z(t)\,dt, \qquad x > 1. \tag{11}$$

By the Fundamental Theorem of Calculus,

$$A'(x) = Z(x). \tag{12}$$

Combining Equation (12) with the sign analysis in Section 4 gives

$$A'(x) < 0, \qquad 1 < x < 2, \tag{13}$$
$$A'(2) = 0, \tag{14}$$
$$A'(x) > 0, \qquad x > 2. \tag{15}$$

Therefore, $A(x)$ decreases on $(1, 2)$ and increases on $(2, \infty)$. The point $x = 2$ is a minimum, not a maximum.

In particular, on the interval $1 < x \leq 200$, the maximum of $A(x)$ occurs at the right endpoint $x = 200$, not at $x = 42$.

Hence the claim that the accumulation function defined by Equation (11) peaks at $2x_p = 42$ is not supported by the stated equations.

A different accumulation measure might have an extremum near 42, but it would need to be defined explicitly and analyzed independently. It would not be valid to attribute such an extremum to Equation (11) without a corresponding derivation.

6 A Candidate Numerical Coefficient

The project also considers the quantity

$$K = 2\pi e^{-1/42}. \tag{16}$$

Numerically,

$$K \approx 6.1353. \tag{17}$$

Equation (16) defines a valid numerical expression. However, the expression does not follow from the definition of $Z(x)$ or from the accumulation function in Equation (11).

The use of $42 = 2x_p$ in Equation (16) is a modeling choice. No result in the present paper establishes K as an asymptotic coefficient associated with the Riemann critical line or the distribution of zeta zeros.

A rigorous investigation of such an interpretation would need to state the target asymptotic relation, derive it from explicit assumptions, and compare it with known results or independent numerical computations.

7 Relation to the Tree-Shadow Study

The companion study, *The Tree and the Shadow*, considers the exact geometric identity

$$\tan\left(\frac{\pi}{8}\right) = \sqrt{2} - 1 \approx 0.41421356. \tag{18}$$

The identity in Equation (18) is exact. Its application to tree crown morphology, however, is an empirical hypothesis rather than a mathematical consequence of trigonometry.

The companion study reports five mango-tree observations with a sample mean crown ratio of approximately 0.41297 and a TOST analysis using an equivalence margin of ± 0.03. Those results are exploratory because the sample is small and comes from a single location.

The numerical proximity between the reported sample mean and $\sqrt{2}-1$ does not establish a causal relationship between tree growth, solar geometry, and the Riemann zeta function.

Similarly, the integer cutoff $x_p = 21$ is defined from $\lfloor \gamma_2 \rfloor$. This definition does not, by itself, establish a mathematical connection between Equation (18) and the zeta zeros.

The proposed relationship among the geometric ratio, biological observations, and numerical cutoff is therefore a research hypothesis requiring independent tests.

8 The Al-Hindara Normalization Proposal

The project uses the relation

$$Z + C + A = 1 \tag{19}$$

as a conceptual statement of unity.

For Equation (19) to become a mathematical identity, the three terms must be defined in a common normalization scheme. The present paper does not supply a derivation showing that the damping function, a separate quantity $C(x)$, and the accumulation function $A(x)$ sum to one.

In particular, setting

$$C(x) = e^{-x/x_p}$$

does not imply that

$$Z(x) + C(x) + A(x) = 1.$$

That equality would have to be established independently, either by construction with explicitly defined normalized quantities or by a proof based on their definitions.

Accordingly, Equation (19) remains a conceptual framework in this paper, not a proven conservation law.

9 Discussion: Evidential Status of the Claims

The claims considered in this paper fall into three categories.

Exact mathematical results

The definition $x_p = \lfloor \gamma_2 \rfloor = 21$ follows from the stated numerical value of γ_2. The sign of $Z(x)$ on the intervals $(1, 2)$ and $(2, \infty)$ follows from the signs of the factors in Equation (3). The derivative relation $A'(x) = Z(x)$ follows from the Fundamental Theorem of Calculus.

Exploratory empirical observations

The companion tree-shadow study reports a small pilot sample whose mean is close to $\sqrt{2} - 1$. This observation is not sufficient to establish a universal biological pattern. Independent replication, transparent measurements, and a suitable sampling design are necessary.

Conjectural modeling proposals

The interpretation of $x_p = 21$ as a meaningful bridge between geometry and number theory, the proposed coefficient K, and the normalization relation $Z + C + A = 1$ require further work. No proof connecting the proposed damping function to the Riemann zeta function is given here.

This separation is necessary to make the framework scientifically testable and to avoid treating numerical coincidence as proof of a mathematical relationship.

10 Conclusion

This paper defines a candidate damping function

$$Z(x) = x^{-5} \ln(x) \sin\left(\frac{2\pi}{x}\right) e^{-x/21}$$

using the integer $21 = \lfloor \gamma_2 \rfloor$.

The function's sign can be determined analytically: it is negative on $1 < x < 2$ and positive on $x > 2$. For the accumulation function

$$A(x) = \int_1^x Z(t)\, dt,$$

this means that $A(x)$ has a minimum at $x = 2$ and is increasing for all $x > 2$. Therefore, the stated model does not support a maximum at $x = 42$.

The quantity $K = 2\pi e^{-1/42}$ is a well-defined numerical expression, but its proposed interpretation as a coefficient related to the Riemann critical line remains unproved.

The companion tree-shadow study supplies an exploratory empirical comparison with the exact value $\sqrt{2} - 1$. That comparison does not establish a causal connection between biological morphology and the zeta zeros.

Future work should focus on defining any alternative accumulation function precisely, testing the model against appropriate null models, providing reproducible numerical code, and proving the normalization relation if it is intended to be an exact mathematical statement.

References

[1] Edwards, H. M. (1974). *Riemann's Zeta Function*. Academic Press.

[2] Schuirmann, D. J. (1987). A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability. *Journal of Pharmacokinetics and Biopharmaceutics*, 15(6), 657–680.

[3] Hallé, F., Oldeman, R. A. A., and Tomlinson, P. B. (1978). *Tropical Trees and Forests: An Architectural Analysis*. Springer-Verlag.

[4] Al-Jabri, A. M. N. (2026). *The Tree and the Shadow: Geometric Derivation of the Reciprocal Silver Ratio and an Exploratory Statistical Analysis of Tree Crown Ratios*. Companion manuscript. Publication status and persistent identifier should be verified before formal submission.

A Reproducibility Checklist

For a reproducible computational study, archive the following:

1. The precise definition and domain of each function.
2. The numerical value used for γ_2 and the operation producing $x_p = 21$.
3. The integration method and numerical tolerance used to evaluate $A(x)$.
4. The search interval and grid resolution for any reported numerical extremum.
5. Code that reproduces the function values, sign analysis, and numerical integration.
6. A clear distinction between numerical observations and analytically proved properties.
7. The definition and derivation of any proposed normalization relation.

B Supplementary Project Resources

The following project resources were supplied by the author. Their current availability and contents should be checked before publication.

- **Zenodo Record (This Paper):**
  https://zenodo.org/records/23540283
- **DOI (This Paper):**
  https://doi.org/10.5281/zenodo.23540283
- Frame 2 visual simulation: https://jabri-com.vercel.app/tree/tree2.html
- Frame 2 TOST tool: https://jabri-com.vercel.app/tree/tree3.html
- Companion manuscript record (Frame 2): https://osf.io/yc9mb/